AQ9.4: Activity Questions 4 - Not Graded

This assignment will not be graded and is only for practice.
Limits:

Level 1:

Find the value of the following limit:

$\lim_{(x,\,y)\,\to\,(2,3)} \frac{x^2+y^2-y}{y^2-x^2}$ lim(x,y)→(2,3) y2−x2x2+y2−y

No, the answer is incorrect.
Score: 0
Accepted Answers:
(Type: Numeric) 2
1 point
1 point
Suppose f f and g g are multivariable functions defined on domain R^2 R2 with the following limits:

$\lim_{(x,\,y)\,\to\,(0,0)} f(x,y) = l$ and $\lim_{(x,\,y)\,\to\,(0,0)} g(x,y) = k$ lim(x,y)→(0,0) f(x, y) = l and lim(x,y)→(0,0) g(x, y) = k

where $l, k \in R$ l, k ∈ R. Which of the following options is(are) correct?

☐ $lim_{(x,y)\to(0,0)}\ (f(x,y)+g(x,y)) = l + k$ lim(x,y)→(0,0) (f(x, y) + g(x, y)) = l + k

☐ $lim_{(x,y)\to(0,0)}\ (f(x,y).g(x,y)) = lk$ lim(x,y)→(0,0) (f(x, y).g(x, y)) = lk

☐ $lim_{(x,y)\to(0,0)}\ (5f(x,y)) \neq 5l$ lim(x,y)→(0,0) (5f(x, y)) = 5l

☐ $lim_{(x,y)\to(0,0)}\ (\frac{f(x,y)}{g(x,y)}) = \frac{l}{k}$ lim(x,y)→(0,0) (g(x,y)f(x,y)) = kl, when $k \neq 0$ k = 0

No, the answer is incorrect.
Score: 0
Accepted Answers:
$lim_{(x,y)\to(0,0)}\ (f(x,y)+g(x,y)) = l + k$ lim(x,y)→(0,0) (f(x, y) + g(x, y)) = l + k

$lim_{(x,y)\to(0,0)}\ (f(x,y).g(x,y)) = lk$ lim(x,y)→(0,0) (f(x, y).g(x, y)) = lk

$lim_{(x,y)\to(0,0)}\ (\frac{f(x,y)}{g(x,y)}) = \frac{l}{k}$ lim(x,y)→(0,0) (g(x,y)f(x,y)) = kl, when $k \neq 0$ k = 0

Consider the function $f: R^2 \longrightarrow R$ f : R2 ⟶ R defined as: $f(x,y) = \begin{cases} \frac{4x^2y^2}{x^2+y^2} & x, y \neq 0 \\ 0 & x = y = 0 \end{cases}$ f(x, y) = {x2+y24x2y20x, y = 0x = y = 0What is the value of $lim_{(x,y)\to(0,0)}\ (f(x,y))$ (x,y)→(0,0)lim (f(x, y))?

No, the answer is incorrect.
Score: 0

Accepted Answers:
(Type: Numeric) 0
1 point
Suppose ff and gg are multivariable functions defined on domain R^2R2 as $f(x, y) = \frac{x^2 + y^2 + y(y - 2x - 1)}{x^2 - y^2}$f(x, y) = x2−y2x2+y2+y(y−2x−1) and $g(x, y) = \frac{2y^2 + x(x - 2y - 1)}{x^2 - y^2}$ g(x, y) = x2−y22y2+x(x−2y−1). Find the value of $lim(x,y)\to(1,1)\ (f(x, y) - g(x, y))$ (x,y)→(1,1)lim (f(x, y) − g(x, y)).

No, the answer is incorrect.
Score: 0
Accepted Answers:
(Type: Numeric) 0.5
1 point
Find the value of the following limit:

$\lim_{(x,\,y)\,\to\,(3,\,1)} \frac{x^2 - 3xy^2}{x^2 - 9y^4}$(x,y)→(3,1)lim x2−9y4x2−3xy2

No, the answer is incorrect.
Score: 0
Accepted Answers:
(Type: Numeric) 0.5
1 point

Level 2:

·· Consider the function $f: R^2 \longrightarrow R$f : R2 ⟶ R defined as:

$$f(x, y) = \begin{cases} \frac{3xy}{x^2 + y^2} & x, y \neq 0 \\ 0 & x = y = 0 \end{cases}$$ f(x, y) = {x2+y23xy0x, y = 0x = y = 0

Use the given information to answer the questions 6-8:

If ff approaches to LL as (x, y)(x, y) approaches to the origin along Y-axis, then find the value of LL?
No, the answer is incorrect.
Score: 0
Accepted Answers:
(Type: Numeric) 0
1 point
If ff approaches to LL as (x, y)(x, y) approaches to the origin along a straight line $y = 3x$y = 3x, then find the value of LL
No, the answer is incorrect.
Score: 0
Accepted Answers:
(Type: Numeric) 0.9
1 point

1 point
Which of the following options is(are) false?
☐

$lim(x,y)\to(0,0)\ \ f(x,y) = 0$(x,y)→(0,0)lim f(x, y) = 0

☐ $lim(x,y)\to(0,0)\ \ f(x,y) = 1$(x,y)→(0,0)lim f(x, y) = 1

☐ Limit of the given function at (0,0) exists but is not equal to 0.
☐ Limit of the given function at (0,0) does not exist.

No, the answer is incorrect.
Score: 0
Accepted Answers:
$lim(x,y)\to(0,0)\ \ f(x,y) = 0$(x,y)→(0,0)lim f(x, y) = 0

$lim(x,y)\to(0,0)\ \ f(x,y) = 1$(x,y)→(0,0)lim f(x, y) = 1

Limit of the given function at (0,0) exists but is not equal to 0.

1 point

Which of the following functions have limit value equal to 5 at the point (1,1,-1)?

☐ $f(x,y,z) = \frac{x+3yz^2-zx}{xy+yz+xz^2}$f(x, y, z) = xy+yz+xz2x+3yz2−zx

☐ $f(x,y,z) = \frac{3(x+y-z)+7x^2y+yz}{xyz+zy+xz}$f(x, y, z) = xyz+zy+xz3(x+y−z)+7x2y+yz

☐ $f(x,y,z) = x^3+y^3+z^3-2xy-2yz-4zx$f(x, y, z) = x3 + y3 + z3 − 2xy − 2yz − 4zx

☐ $f(x,y,z) = \frac{x^2e^{\frac{x}{2}}+y^2e^{\frac{y}{2}}+z^2e^{\frac{-z}{2}}}{e^{\frac{x+y+z}{2}}}$f(x, y, z) = e2x+y+z x2e2x+y2e2y+z2e2−z

No, the answer is incorrect.
Score: 0
Accepted Answers:
$f(x,y,z) = \frac{x+3yz^2-zx}{xy+yz+xz^2}$f(x, y, z) = xy+yz+xz2x+3yz2−zx
$f(x,y,z) = x^3+y^3+z^3-2xy-2yz-4zx$f(x, y, z) = x3 + y3 + z3 − 2xy − 2yz − 4zx

1 point

Consider the function $f: R^2 \longrightarrow R$f : R2 ⟶ R defined as:

$$f(x,y) = \begin{cases} \frac{x^k y}{x^{2n}+y^{2n}} & x, y \neq 0 \\ 0 & x = y = 0 \end{cases}$$f(x, y) = {x2n+y2nxkyx, y ≠ 0 0x = y = 0

where $k, n \in N \setminus \{0\}$k, n ∈ N \ {0}. Which of the following statements is(are) true about ff?

☐ If $k = 2n-1$k = 2n − 1, then the limit at (0,0) exists and is equal to 0.

☐ If $k < 2n-1$k < 2n − 1, then the limit at (0,0) does not exist.

☐ If $k > 2n$k > 2n, then the limit at (0,0) always exists and is equal to 0.

☐ If $k > 2n$k > 2n, then the limit at (0,0) does not exist.

No, the answer is incorrect.
Score: 0
Accepted Answers:
If $k < 2n-1$k < 2n − 1, then the limit at (0,0) does not exist.
If $k > 2n$k > 2n, then the limit at (0,0) always exists and is equal to 0.

Check Answers

Your score is: 0/10